

# Disproof of Conjecture 00000003831: a seminormal $\mathfrak{gl}_3$ crystal whose character is $s_{21} - s_{111}$

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## Abstract

Conjecture 00000003831 asserts that for every finite crystal  $B$  the Schur expansion of the Weyl symmetrization of the character  $\text{ch}(B)$  has nonnegative coefficients, and that for  $B = B(\lambda) \otimes B(\mu)$  these coefficients are the Littlewood–Richardson coefficients. The first clause is false. We exhibit a connected 7-vertex *seminormal*  $\mathfrak{gl}_3$  crystal  $B_7$ , obtained from  $B(2, 1, 0)$  by identifying its two vertices of weight  $(1, 1, 1)$  into one vertex and keeping all eight edges. It satisfies all of Kashiwara’s crystal axioms,  $\varepsilon_i, \varphi_i$  are the string lengths, and

$$\text{ch}(B_7) = m_{21} + m_{111} = s_{21} - s_{111}.$$

The character is symmetric, so the Weyl symmetrizer  $J(x^\rho f)/J(x^\rho)$  fixes it and the coefficient of  $s_{111}$  is  $-1$ ; the orbit sum gives  $6s_{21} - 6s_{111}$  and the orbit average gives  $s_{21} - s_{111}$ . Seven vertices is the minimum for a seminormal  $\mathfrak{gl}_3$  counterexample, and seminormal  $\mathfrak{gl}_2$  crystals never give one. Clause 1 is true for *normal* (Stembridge) crystals, so it can only be rescued by normal-type readings of “finite crystal” (normal crystals, or the affine notion of a finite crystal, which is classically normal). The refutation is formalised in Lean 4 (core library only): crystals, Kashiwara’s axioms, seminormality, characters, Schur polynomials (from tableaux), the antisymmetrizer and the Weyl symmetrization are defined from scratch. An independent Python script rechecks everything and performs the minimality searches.

## 1 The claim

The conjecture (English version): “*Definition: the crystal character  $\text{ch}(B)$  denotes the generating function of vertex weights. Conjecture: for any finite crystal, the coefficients of the Schur expansion of the Weyl symmetrization of  $\text{ch}(B)$  are nonnegative, and for  $B = B(\lambda) \otimes B(\mu)$  these coefficients are exactly the classical Littlewood–Richardson coefficients.*”

It is a conjunction of two clauses:

- (C1) for every finite crystal  $B$ , the Schur expansion of the Weyl symmetrization of  $\text{ch}(B)$  has nonnegative coefficients;
- (C2) for  $B = B(\lambda) \otimes B(\mu)$  these coefficients are the LR coefficients  $c_{\lambda\mu}^\nu$ .

We refute (C1); one false clause makes the conjunction false. Under the Weyl-symmetrizer reading (C2) is *true*:  $\text{ch}(B(\lambda) \otimes B(\mu)) = s_\lambda s_\mu$  is symmetric, so it is fixed by the symmetrizer, and  $s_\lambda s_\mu = \sum_\nu c_{\lambda\mu}^\nu s_\nu$ . The disproof rests entirely on (C1).

## 2 Definitions

**Crystals.** We work with  $\mathfrak{gl}_3$ : weight lattice  $P = \mathbb{Z}^3$  with basis  $\epsilon_1, \epsilon_2, \epsilon_3$ , simple roots  $\alpha_1 = \epsilon_1 - \epsilon_2$ ,  $\alpha_2 = \epsilon_2 - \epsilon_3$ , and coroots acting by  $\langle h_1, w \rangle = w_1 - w_2$ ,  $\langle h_2, w \rangle = w_2 - w_3$ . Following Kashiwara [2, 3] and Bump–Schilling [1, Ch. 2], a *crystal* is a set  $B$  with maps  $\text{wt}: B \rightarrow P$ ,  $\varepsilon_i, \varphi_i: B \rightarrow \mathbb{Z} \sqcup \{-\infty\}$  and  $\tilde{e}_i, \tilde{f}_i: B \rightarrow B \sqcup \{0\}$  ( $i = 1, 2$ ) such that

1.  $\varphi_i(b) = \varepsilon_i(b) + \langle h_i, \text{wt}(b) \rangle$ ;
2.  $\tilde{f}_i b = b'$  if and only if  $\tilde{e}_i b' = b$ ;
3. if  $\tilde{e}_i b \in B$  then  $\text{wt}(\tilde{e}_i b) = \text{wt}(b) + \alpha_i$ ,  $\varepsilon_i(\tilde{e}_i b) = \varepsilon_i(b) - 1$ ,  $\varphi_i(\tilde{e}_i b) = \varphi_i(b) + 1$ ;
4. if  $\tilde{f}_i b \in B$  then  $\text{wt}(\tilde{f}_i b) = \text{wt}(b) - \alpha_i$ ,  $\varepsilon_i(\tilde{f}_i b) = \varepsilon_i(b) + 1$ ,  $\varphi_i(\tilde{f}_i b) = \varphi_i(b) - 1$ ;
5. if  $\varphi_i(b) = -\infty$  then  $\tilde{e}_i b = \tilde{f}_i b = 0$ .

A crystal is *seminormal* if  $\varepsilon_i(b) = \max\{k \geq 0 : \tilde{e}_i^k b \neq 0\}$  and  $\varphi_i(b) = \max\{k \geq 0 : \tilde{f}_i^k b \neq 0\}$  for all  $i, b$ ; it is *normal* if it is (isomorphic to) the crystal of an integrable  $U_q(\mathfrak{gl}_3)$ -module, equivalently a disjoint union of highest-weight crystals  $B(\lambda)$  (for simply-laced types this is equivalent to Stembridge’s local axioms [4]). In these references seminormality and normality are *extra* conditions; “crystal” alone means the abstract notion above.

**Characters and Schur polynomials.** For a finite crystal,  $\text{ch}(B) = \sum_{b \in B} x^{\text{wt}(b)} \in \mathbb{Z}[x_1^{\pm 1}, x_2^{\pm 1}, x_3^{\pm 1}]$ . The Weyl group  $S_3$  permutes the variables. Let  $J(f) = \sum_{w \in S_3} \text{sgn}(w) w(f)$ ,  $\rho = (2, 1, 0)$  and  $a_\mu = J(x^\mu)$ . For a dominant weight  $\lambda$  ( $\lambda_1 \geq \lambda_2 \geq \lambda_3$ , integers) the Schur polynomial is  $s_\lambda = a_{\lambda+\rho}/a_\rho$ ; for a partition  $\lambda$  it equals  $\sum_T x^T$  over semistandard Young tableaux of shape  $\lambda$  with entries in  $\{1, 2, 3\}$ , and  $s_{\lambda+(k,k,k)} = (x_1 x_2 x_3)^k s_\lambda$ .

**Lemma 1** (uniqueness of Schur expansions). *The  $s_\lambda$ ,  $\lambda$  dominant, are linearly independent, and a symmetric Laurent polynomial  $f$  equals  $\sum_\lambda c_\lambda s_\lambda$  with  $c_\lambda = [x^{\lambda+\rho}] J(x^\rho f)$ .*

*Proof.* If  $f$  is symmetric then  $J(x^\rho f) = f a_\rho$ , an antisymmetric Laurent polynomial. An antisymmetric Laurent polynomial is determined by its coefficients on the strictly decreasing exponents  $\mu_1 > \mu_2 > \mu_3$  (every exponent with two equal entries has coefficient 0, and the others are  $\pm$  a strictly decreasing one), so  $J(x^\rho f) = \sum_\lambda c_\lambda a_{\lambda+\rho}$  with  $c_\lambda$  as stated. Dividing by  $a_\rho$  gives the expansion. If  $\sum c_\lambda s_\lambda = 0$  then  $\sum c_\lambda a_{\lambda+\rho} = 0$ , and the  $a_\mu$  with distinct strictly decreasing  $\mu$  have disjoint supports, so all  $c_\lambda = 0$ .  $\square$

**Readings of “Weyl symmetrization”.** For a Laurent polynomial  $f$ :

- (W) the Weyl symmetrizer  $\pi(f) = J(x^\rho f)/J(x^\rho)$  (the Weyl character formula operator, equal to the Demazure operator  $\pi_{w_0} = \pi_1 \pi_2 \pi_1$ ); its Schur coefficients are  $[x^{\lambda+\rho}] J(x^\rho f)$  for any  $f$ ;
- (O) the orbit sum  $\sum_{w \in S_3} w(f)$ ;
- (A) the orbit average  $\frac{1}{6} \sum_{w \in S_3} w(f)$ .

If  $f$  is symmetric then  $\pi(f) = f$  (because  $J(x^\rho f) = f J(x^\rho)$ ), the orbit sum is  $6f$  and the average is  $f$ .

### 3 The counterexample

The crystal  $B_7$  has seven vertices

$$A(2, 1, 0), B(1, 2, 0), C(2, 0, 1), D(0, 2, 1), E(1, 0, 2), F(0, 1, 2), Z(1, 1, 1)$$

(weights in brackets), and the edges

$$\tilde{f}_1 : A \rightarrow B, \quad C \rightarrow Z \rightarrow D, \quad E \rightarrow F; \quad \tilde{f}_2 : A \rightarrow C, \quad B \rightarrow Z \rightarrow E, \quad D \rightarrow F,$$

with  $\tilde{e}_i$  the inverse partial maps and  $\tilde{e}_i, \tilde{f}_i = 0$  elsewhere. The values of  $\varepsilon_i, \varphi_i$  are the string lengths:

|                              | $A$       | $B$       | $C$       | $D$       | $E$       | $F$       | $Z$       |
|------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| wt                           | (2, 1, 0) | (1, 2, 0) | (2, 0, 1) | (0, 2, 1) | (1, 0, 2) | (0, 1, 2) | (1, 1, 1) |
| $(\varepsilon_1, \varphi_1)$ | (0, 1)    | (1, 0)    | (0, 2)    | (2, 0)    | (0, 1)    | (1, 0)    | (1, 1)    |
| $(\varepsilon_2, \varphi_2)$ | (0, 1)    | (0, 2)    | (1, 0)    | (0, 1)    | (2, 0)    | (1, 0)    | (1, 1)    |

The crystal  $B(2, 1, 0)$  has the eight tableaux  $11/2, 12/2, 11/3, 12/3, 13/2, 13/3, 22/3, 23/3$  (written first row/second row), with  $\tilde{f}_1$ -edges  $11/2 \rightarrow 12/2, 11/3 \rightarrow 12/3 \rightarrow 22/3, 13/3 \rightarrow 23/3$  and  $\tilde{f}_2$ -edges  $11/2 \rightarrow 11/3, 12/2 \rightarrow 13/2 \rightarrow 13/3, 22/3 \rightarrow 23/3$ . The two vertices of weight  $(1, 1, 1)$  are  $12/3$  (on a 1-string of length 2, alone in colour 2) and  $13/2$  (alone in colour 1, on a 2-string of length 2). Identifying them to a single vertex  $Z$  keeps all eight edges and gives exactly  $B_7$ .

**Proposition 2.**  $B_7$  is a connected seminormal crystal (with all  $\varepsilon_i, \varphi_i$  finite).

*Proof.* Each  $\tilde{f}_i$  is injective on its domain, so  $\tilde{e}_i$  is a well-defined inverse and axiom 2 holds. Each edge  $u \rightarrow v$  of colour  $i$  has  $\text{wt}(u) - \text{wt}(v) = \alpha_i$  (check the four edges of each colour, e.g.  $C \rightarrow Z$ :  $(2, 0, 1) - (1, 1, 1) = (1, -1, 0) = \alpha_1$ ,  $Z \rightarrow E$ :  $(1, 1, 1) - (1, 0, 2) = (0, 1, -1) = \alpha_2$ ). The  $i$ -strings are  $A \rightarrow B, C \rightarrow Z \rightarrow D, E \rightarrow F$  for  $i = 1$  and  $A \rightarrow C, B \rightarrow Z \rightarrow E, D \rightarrow F$  for  $i = 2$ ; every vertex lies on exactly one string of each colour. Setting  $\varepsilon_i(b)$  (resp.  $\varphi_i(b)$ ) to the number of steps from  $b$  to the top (resp. bottom) of its string gives the table above, makes  $B_7$  seminormal by definition, and gives axioms 3 and 4 automatically. Axiom 1 is a direct check: for a string  $b_0 \rightarrow \dots \rightarrow b_\ell$  of colour  $i$  we have  $\langle h_i, \text{wt}(b_j) \rangle = \langle h_i, \text{wt}(b_0) \rangle - 2j$ , and from the table  $\langle h_i, \text{wt}(b_0) \rangle = \ell$  in all six strings (for instance  $\langle h_1, \text{wt}(C) \rangle = 2 - 0 = 2$ ,  $\langle h_2, \text{wt}(B) \rangle = 2 - 0 = 2$ ,  $\langle h_1, \text{wt}(A) \rangle = 1$ ). Hence  $\varphi_i(b_j) - \varepsilon_i(b_j) = (\ell - j) - j = \langle h_i, \text{wt}(b_j) \rangle$ . Axiom 5 is void. Connectedness: every vertex is joined to  $A$ .  $\square$

**Proposition 3.**  $\text{ch}(B_7) = s_{(2,1,0)} - s_{(1,1,1)}$ , and this is the unique Schur expansion.

*Proof.* The weights of  $B_7$  are the six permutations of  $(2, 1, 0)$  and  $(1, 1, 1)$ , each once, so  $\text{ch}(B_7) = m_{21} + m_{111}$ , which is symmetric. The semistandard tableaux of shape  $(2, 1)$  with entries  $\leq 3$  are the eight listed above, so  $s_{21} = m_{21} + 2m_{111}$ , and  $s_{111} = x_1 x_2 x_3 = m_{111}$ . Hence  $\text{ch}(B_7) = s_{21} - s_{111}$ . Uniqueness is Lemma 1; in degree 3 it is also visible directly, since the coefficients of  $x^{300}, x^{210}, x^{111}$  in  $s_{300}, s_{210}, s_{111}$  form the unitriangular matrix  $\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 2 & 1 \end{pmatrix}$ .  $\square$

**Theorem 4.** Clause (C1) fails for  $B_7$  under each of the readings  $(W), (O), (A)$ :

$$\pi(\text{ch } B_7) = s_{21} - s_{111}, \quad \sum_w w(\text{ch } B_7) = 6s_{21} - 6s_{111}, \quad \frac{1}{6} \sum_w w(\text{ch } B_7) = s_{21} - s_{111}.$$

The coefficient of  $s_{111}$  is  $-1, -6, -1$  respectively. Consequently the conjecture is false.

*Proof.*  $\text{ch } B_7$  is symmetric, so by the remark after the readings the three symmetrizations are  $f, 6f, f$  with  $f = \text{ch } B_7$ ; Proposition 3 and Lemma 1 give the expansions. Explicitly,  $J(x^\rho \text{ch } B_7) = a_{(4,2,0)} - a_{(3,2,1)}$ , so  $[x^{(1,1,1)+\rho}]J(x^\rho \text{ch } B_7) = -1$ .  $\square$

## 4 What is true, and the boundary of the counterexample

**Normal crystals.** If  $B$  is normal then  $B \cong \bigsqcup_k B(\lambda^{(k)})$  and  $\text{ch } B = \sum_k s_{\lambda^{(k)}}$  (the character of  $B(\lambda)$  is the irreducible character  $s_\lambda$ ), which is symmetric and Schur-nonnegative; all three readings then give nonnegative coefficients. In particular (C1) holds for  $B(\lambda) \otimes B(\mu)$ , and (C2) holds under (W) and (A); under (O) the coefficients are  $6c_{\lambda\mu}^\nu$ , so (C2) fails under (O). Our  $B_7$  is not normal: a normal crystal has a Schur-positive character. So the only readings under which (C1) survives are normal-type ones: “finite crystal” = “normal crystal”, or the affine-theory term discussed in Section 5. The literal statement says “any finite crystal”, and in the standard references [2, 3, 1] “crystal” does not include normality (or even seminormality);  $B_7$  is moreover seminormal, the strongest condition short of normality that is usually attached to the word.

### Characters of seminormal crystals.

**Lemma 5.** *Let  $B$  be a finite seminormal  $\mathfrak{gl}_r$  crystal and  $m(\nu)$  the number of vertices of weight  $\nu$ . Then  $m$  is  $S_r$ -invariant, and for every  $i$  and every  $\nu$  with  $\langle h_i, \nu \rangle \geq 0$  we have  $m(\nu + \alpha_i) \leq m(\nu)$  (the string condition). Conversely every finitely supported  $m \geq 0$  with these two properties is the weight multiplicity function of a finite seminormal crystal.*

*Proof.* Seminormality makes every vertex lie on a unique  $i$ -string  $b_0 \rightarrow \cdots \rightarrow b_\ell$  with  $\varepsilon_i(b_0) = 0$ ,  $\varphi_i(b_0) = \ell$ , so  $\langle h_i, \text{wt } b_0 \rangle = \ell$  and the weights  $\text{wt } b_0 - j\alpha_i$ ,  $0 \leq j \leq \ell$ , are permuted by the reflection  $s_i$ . Hence  $m$  is  $s_i$ -invariant for all  $i$ , i.e.  $S_r$ -invariant. If  $\langle h_i, \nu \rangle = k \geq 0$ , a string through  $\nu + \alpha_i$  contains  $s_i(\nu + \alpha_i) = \nu - (k+1)\alpha_i$  and is contiguous, so it contains  $\nu$ ; this gives  $m(\nu + \alpha_i) \leq m(\nu)$ . Conversely, for each colour separately, cut every line  $\nu + \mathbb{Z}\alpha_i$  into strings by the usual layering (the multiplicities along the line are symmetric about the reflecting point and weakly decrease away from it), and define  $\tilde{f}_i, \tilde{e}_i, \varepsilon_i, \varphi_i$  along these strings. The axioms only involve one colour at a time, so the colours do not interact.  $\square$

**Proposition 6** ( $\mathfrak{gl}_2$ ). *Every finite seminormal  $\mathfrak{gl}_2$  (or  $\mathfrak{sl}_2$ ) crystal has a Schur-positive character; so no counterexample exists for  $\mathfrak{gl}_2$  among seminormal crystals.*

*Proof.* There is one colour;  $B$  is the disjoint union of its strings, and a string with top weight  $(a, c)$  has  $\ell = a - c \geq 0$  and character  $\sum_{j=0}^{\ell} x_1^{a-j} x_2^{c+j} = s_{(a,c)}$ .  $\square$

For  $\mathfrak{gl}_3$  each colour still decomposes  $\text{ch } B$  into  $\mathfrak{sl}_2$ -characters, but the two decompositions need not come from a common  $\mathfrak{gl}_3$ -structure;  $B_7$  exploits exactly this.

**Proposition 7** (minimality). *Every finite seminormal  $\mathfrak{gl}_3$  crystal with at most 6 vertices has a Schur-positive character. Hence 7 is the minimum number of vertices of a seminormal  $\mathfrak{gl}_3$  counterexample to (C1).*

*Proof.* Edges preserve the degree  $w_1 + w_2 + w_3$ , and the Schur expansion is compatible with the decomposition by degree, so a minimal counterexample lives in one degree  $d$ ; multiplying by  $(x_1 x_2 x_3)^k$  maps  $s_\lambda$  to  $s_{\lambda + (k,k,k)}$  and preserves everything, so we may take  $d \in \{0, 1, 2\}$ . If a dominant  $\lambda$  occurs,  $p = \lambda_1 - \lambda_2$ , then by Lemma 5 the  $p+1$  distinct weights  $\lambda - j\alpha_1$  ( $0 \leq j \leq p$ ) all occur, so  $p \leq N-1$ , and likewise  $\lambda_2 - \lambda_3 \leq N-1$ , where  $N$  is the number of vertices. This leaves a finite list of multiplicity functions; `verify.py` enumerates all of them for  $N \leq 6$ , keeps those satisfying the conditions of Lemma 5 (necessary conditions suffice here) and finds exactly 12, namely  $k s_{000}$  ( $1 \leq k \leq 6$ ),  $s_{100}$ ,  $2s_{100}$ ,  $s_{110}$ ,  $2s_{110}$ ,  $s_{200}$  and  $s_{(1,1,-1)}$ , all Schur-positive. For  $N = 7$  there is exactly one non-positive admissible character up to the shift,  $m_{(1,0,-1)} + m_{000} = s_{(1,0,-1)} - s_{000}$ , which is  $\text{ch}(B_7)$  shifted by  $(x_1 x_2 x_3)^{-1}$ ; the script also builds a seminormal crystal for it and rechecks the axioms.  $\square$

*Remark 8* (a cheaper abstract example). Without seminormality, one vertex suffices: the  $\mathfrak{gl}_2$  crystal  $\{b\}$  with  $\text{wt}(b) = (-1, 2)$ ,  $\varepsilon(b) = 3$ ,  $\varphi(b) = 0$ ,  $\tilde{e}b = \tilde{f}b = 0$  satisfies Kashiwara’s axioms ( $\varphi - \varepsilon = -3 = \langle h, \text{wt } b \rangle$ ). Then  $J(x^\rho x_1^{-1} x_2^2) = a_{(0,2)} = -a_{(2,0)}$ , so the Weyl symmetrization is  $-s_{(1,0)}$ , and the orbit sum is  $s_{(2,-1)} - s_{(1,0)}$ . A second one: the lowest vertex  $b$  of  $B(2, 0)$ , kept alone with its inherited values  $\text{wt}(b) = (0, 2)$ ,  $\varepsilon(b) = 2$ ,  $\varphi(b) = 0$  and no edges, is a crystal ( $\varphi - \varepsilon = -2 = 0 - 2$ ). Its character is  $x_2^2$ , and  $J(x^\rho x_2^2) = x_1 x_2^2 - x_1^2 x_2 = -a_{(2,1)}$ , so the Weyl symmetrization is  $-s_{(1,1)}$  (orbit sum  $x_1^2 + x_2^2 = s_{(2,0)} - s_{(1,1)}$ ). We do not rely on these examples: it is far from the spirit of the conjecture, while  $B_7$  is seminormal, connected and differs from  $B(2, 1, 0)$  only by one vertex.

## 5 Readings covered

- *Crystal*: abstract Kashiwara crystals and seminormal crystals ( $B_7$  is seminormal). Only normal-type readings rescue (C1).
- *The symmetrization would be idle for normal crystals*. For normal (indeed seminormal, Lemma 5) crystals  $\text{ch}(B)$  is already  $W$ -invariant, so “Weyl symmetrization” would be vacuous, and for normal crystals (C1) would be a triviality. The explicit symmetrization therefore points to the abstract notion, under which even a 1-vertex crystal is a counterexample (the remark above).  $B_7$  refutes (C1) even under the stronger seminormal reading.
- *Affine reading*. In affine crystal theory, a “finite crystal” means the crystal base of a finite-dimensional  $U'_q(\hat{\mathfrak{g}})$ -module (e.g. Kirillov–Reshetikhin or perfect crystals). Such crystals are classically normal: their restriction to the classical subalgebra is a disjoint union of  $B(\lambda)$ ’s, so (C1) holds under this reading. It is one of the normal-type readings named above, and our counterexample does not apply to it.
- *Weyl symmetrization*: the symmetrizer  $J(x^\rho f)/J(x^\rho)$  (Demazure operator  $\pi_{w_0}$ ), its bialternant coefficients  $[x^{\lambda+\rho}]J(x^\rho f)$ , the orbit sum and the orbit average all give a negative coefficient. Since  $\text{ch } B_7$  is symmetric, any symmetrization that fixes symmetric functions, or multiplies them by a positive constant, does.
- $\mathfrak{gl}_3$  versus  $\mathfrak{sl}_3$ : projecting the weights to the  $\mathfrak{sl}_3$  weight lattice turns  $s_\lambda$  into the irreducible character  $\chi_{\bar{\lambda}}$ , and  $\text{ch } B_7 = \chi_{(1,1)} - \chi_{(0,0)}$  (in fundamental-weight coordinates); the coefficient of the trivial character is  $-1$ .
- *Polynomial versus Laurent*: all weights of  $B_7$  are nonnegative, so  $\text{ch } B_7$  is an honest polynomial and the expansion is in ordinary Schur polynomials.

## 6 Machine verification

**Lean 4** (`lean4/Main.lean`, Lean 4.19.0, core library only, no `sorry`, no `native.decide`; the main theorems use only `propext` and `Quot.sound`, some lemmas also `Classical.choice`). Everything is defined from scratch:

- **Crystal n**:  $\text{wt}: \text{Fin } n \rightarrow \mathbb{Z}^3$ ,  $\tilde{e}_i, \tilde{f}_i: \text{Fin } n \rightarrow \text{Option } (\text{Fin } n)$ ,  $\varepsilon_i, \varphi_i: \text{Fin } n \rightarrow \mathbb{Z}$ ; `IsCrystal` (axioms 1–4) and `IsSeminormal`; `seminormal_eps_max` and `seminormal_phi_max` prove that it means  $\varepsilon_i = \max\{k : \tilde{e}_i^k b \neq 0\}$  (and likewise for  $\varphi_i$ ).

- Laurent polynomials as finite formal sums with coefficient-wise equality `PolyEq` (a sound and complete decision procedure is proved), multiplication, the  $S_3$ -action,  $J$ , the orbit sum, the character `ch`.
- `schur`: Schur polynomials as sums over semistandard tableaux (all fillings of the shape with entries in  $\{1, 2, 3\}$  are generated and filtered); the bialternant identities  $s_\lambda a_\rho = a_{\lambda+\rho}$  for  $\lambda \vdash 3$ ; `schur3_indep` (uniqueness of degree-3 Schur expansions).
- `WeylSym f q`:  $q \cdot J(x^\rho) = J(x^\rho f)$ ; `schurCoeffW`:  $[x^{\lambda+\rho}]J(x^\rho f)$ .
- `B7_isCrystal`, `B7_seminormal`, `ch_B7_expands` (`ch B7 = s210 - s111`), `weylSym_B7`, `weyl_forced` (for every  $q$  with `WeylSym (ch B7) q` and every degree-3 expansion of  $q$  the coefficients are  $(0, 1, -1)$ ), `orbit_forced`, `avg_forced`, `B7_counterexample`.
- Main theorems: `conjecture_00000003831_false`:  $\neg \text{ClaimWeyl}$  and the variants `_abstract`, `_balt`, `_orbit`, `_avg` for the other readings.
- Non-vacuity: `B100_ok` ( $B(1, 0, 0)$  is a seminormal crystal with character  $s_{100}$ ), `B210_ok` ( $B(2, 1, 0)$  satisfies all hypotheses and has expansion  $s_{210}$ , nonnegative), `broken_rejected` (a datum violating axiom 1 is rejected).

Limitations: in Lean the Schur expansions quantified over are the degree-3 ones  $as_{300} + bs_{210} + cs_{111}$  (sufficient because `ch B7` is homogeneous of degree 3; general uniqueness is Lemma 1). That the symmetrizer fixes symmetric polynomials is checked for `ch B7`, not proved in general. Normality, Lemma 5, the  $\mathfrak{gl}_2$  statement and minimality are in this report and `verify.py` only.

**Python** (`verify.py`, standard library). It checks the axioms by a different method (string lengths computed by walking), computes Schur polynomials both from tableaux and by exact division  $a_{\lambda+\rho}/a_\rho$ , computes the symmetrizer by exact polynomial division and via Demazure operators  $\pi_1\pi_2\pi_1$ , the orbit sum and average (with fractions), runs the  $\mathfrak{gl}_2$  (up to 10 vertices) and  $\mathfrak{gl}_3$  (up to 7 vertices) searches of Proposition 7, and sanity-checks the normal crystals  $B(1, 0, 0)$ ,  $B(2, 1, 0)$  and  $B(1)^{\otimes 3}$  (tensor rule).

## References

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